

SAT Math Mastery · Week TP1E**SAT Trick Pack 1 — Part E****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** statistics · exponential shifts · perpendicular lines · maximum vs. supremum · composite unknowns · inradius · equivalent quadratics · exponent rewriting · integer statements · circles · percent change · identities · solids · two scenarios · pyramids**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Decide how removing values changes the mean and the spread without recomputing.
- S2.** Remove an unknown vertical shift from an exponential so the outputs form a geometric sequence.
- S3.** Build a perpendicular line symbolically: negative reciprocal slope plus a known point.
- S4.** Tell an attained maximum from a bound that is only approached.
- S5.** Treat a product of unknowns as one variable and eliminate.
- S6.** Use $r = \frac{a+b-c}{2}$ for the inradius of a right triangle with special angles.
- S7.** Match equivalent quadratic forms: leading coefficient, axis, then one root.
- S8.** Factor a composite base into primes and shift exponents so the constant appears as a coefficient.
- S9.** Test “must be true” integer statements by structure or a single counterexample.
- S10.** Substitute a centre into another circle’s equation; use symmetry of a chord in general form.
- S11.** Track component percent changes from a baseline of 100 and compute the new share.
- S12.** Clear denominators and match coefficients of an identity.
- S13.** Hollow solids: outer minus inner with the difference of squares; reject the impossible root.
- S14.** Eliminate a shared unknown between two scenarios; a rectangular pyramid has two slant heights.

2 · REQUIRED FACTS

- Removing two values whose average equals the mean leaves the mean unchanged; removing the extremes lowers the standard deviation.
- For $y = ab^x + c$ at equally spaced inputs, $y - c$ is a geometric sequence, so $(y_2 - c)^2 = (y_1 - c)(y_3 - c)$.
- Perpendicular slopes multiply to -1 .
- A decaying exponential never reaches its horizontal bound; a downward parabola attains its vertex value.
- Right-triangle inradius: $r = \frac{a+b-c}{2}$ (legs a, b , hypotenuse c).
- If $p(x) = a(x-r)(x-s)$ then the axis is $x = \frac{r+s}{2}$ and $p(r) = 0$.
- For $(x+p)(x+q) = x^2 + bx + c$: $p+q = -b$, $pq = c$, $p^2 + q^2 = b^2 - 2c$.
- A circle’s equation is satisfied by every point on it; two points with equal y put the centre on the vertical line through their midpoint.
- A right pyramid on an $l \times w$ base has two slant heights, $\sqrt{h^2 + (w/2)^2}$ and $\sqrt{h^2 + (l/2)^2}$.

3 · REQUIRED FORMULAS

$$(y_2 - c)^2 = (y_1 - c)(y_3 - c), \quad m_1 m_2 = -1, \quad r = \frac{a+b-c}{2}, \quad a^2 - b^2 = (a-b)(a+b)$$

$$V_{\text{hollow}} = \pi h (R^2 - r^2), \quad P = \frac{F}{A}, \quad \text{share} = \frac{\text{part}}{\text{total}}, \quad S_{\text{pyramid}} = lw + l s_1 + w s_2$$

4 · MINI THEORY SHEET

Shifts and parameters. If an unknown constant c is added to ab^x , do not solve for a and b first: subtract c from each output, apply the geometric-mean relation, and the unknowns a, b disappear.

Percent change. Assume a baseline total of 100, write each component as a multiplier, add to get the new total, and divide by the *new* total to get a share.

Equivalent forms. Match the leading coefficients, then use whichever of axis, root and vertex is cheapest; never expand both sides.

Two scenarios. Write the same relation twice. A quantity that is the same in both (a weight, a total) can be eliminated before anything else is solved.

Geometry with a parameter. Substitute the known point or centre; clear squares early; check the answer against any location condition.

Speed shortcuts.

Situation	Shortcut
remove the min and the max	check whether they average to the mean; spread falls
$ab^x + c$ at equal spacing	$(y_2 - c)^2 = (y_1 - c)(y_3 - c)$
perpendicular to a line with parameter	negate and invert the slope, point-slope form
“greatest value” of a decay model	the horizontal bound is not attained; a parabola’s vertex is
product such as ay in two equations	call it one unknown and eliminate
right triangle with incircle	$r = \frac{a+b-c}{2}$ with the special ratios
vertex form = factored form	equal leading coefficients, axis is the midpoint of roots, plug a root
12^{x+k} with an unknown multiplier	$12^k \cdot 12^x$ and split $12 = 2^2 \cdot 3$
“must be true” about integers	use $b^2 - 2c$ type identities, or one counterexample
centre of one circle on another	substitute; pull square factors out early
percent changes by component	baseline 100; new share = part $\div$ new total
fractions equal for all x	clear denominators; compare coefficients of each power
hollow cylinder	$R^2 - r^2 = (R - r)(R + r)$; keep the root with $0 < t < R$
same weight, two arrangements	solve the weight from one, substitute into the other
rectangular pyramid	two slant heights; two pairs of faces

5 · GUIDED EXAMPLES

Example 1 — Remove the shift first

$f(x) = ab^x + c$ with $f(0) = 5$, $f(1) = 7$, $f(2) = 11$. Find c .

$f - c$ is geometric: $(7 - c)^2 = (5 - c)(11 - c) \Rightarrow 49 - 14c = 55 - 16c \Rightarrow c = 3$.

Check: 2, 4, 8 has ratio 2.

Example 2 — Inradius of a right triangle

A right isosceles triangle has inradius 2. Find its hypotenuse.

Legs s , hypotenuse $s\sqrt{2}$: $r = \frac{2s - s\sqrt{2}}{2} = 2 \Rightarrow s = \frac{4}{2 - \sqrt{2}} = 4 + 2\sqrt{2}$.

Hypotenuse = $s\sqrt{2} = 4\sqrt{2} + 4$.

Example 3 — Baseline of 100 and the new share

Electronics rise 40%, everything else falls 10%, the total rises 5%. Original electronics share?

$1.4x + 0.9(100 - x) = 105 \Rightarrow 0.5x = 15 \Rightarrow x = 30$. New electronics = 42, new total = 105, so the new share is $\frac{42}{105} = 40\%$.

6 · PRACTICE SET — 16 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

A data set has 10 values: 31, 38, 42, 45, 47, 52, 53, 58, 65, 69. The mean of the data set is 50. The least value and the greatest value are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

(A) Mean greater; SD smaller

(B) Mean unchanged; SD smaller ✓

(C) Mean unchanged; SD unchanged

(D) Mean smaller; SD smaller

- The removed values sum to $31 + 69 = 100$, so they average 50, exactly the mean. Removing them leaves the mean unchanged: the remaining sum is $500 - 100 = 400$ and $\frac{400}{8} = 50$.
- The two removed values are the farthest from the mean (19 below and 19 above), so every deviation that remains is smaller than 19 in size. The sum of squared deviations drops sharply while 8 values remain, so the standard deviation decreases.
- (Numerically, the population standard deviation falls from about 11.3 to about 8.2.)

Answer: (B)

2.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = ab^x + c$, where a , b , and c are constants and $b > 1$. The values $f(2) = 14$, $f(4) = 20$, and $f(6) = 32$ are known. What is the value of c ?

- The inputs 2, 4, 6 are equally spaced, so $f(2) - c$, $f(4) - c$, $f(6) - c$ form a geometric sequence: $(20 - c)^2 = (14 - c)(32 - c)$.
- $400 - 40c + c^2 = 448 - 46c + c^2 \Rightarrow 6c = 48 \Rightarrow c = 8$.
- Check: 6, 12, 24 has constant ratio $2 = b^2$, so $b = \sqrt{2}$ and $a = 3$; indeed $3 \cdot 2 + 8 = 14$, $3 \cdot 4 + 8 = 20$, $3 \cdot 8 + 8 = 32$.

Answer: 8

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

Line p has equation $3x - ky = 15$, where k is a nonzero constant. Line q is perpendicular to line p and passes through the y -intercept of line p . Which of the following is an equation of line q ?

(A) $y = \frac{k}{3}x - \frac{15}{k}$

(B) $y = -\frac{3}{k}x - \frac{15}{k}$

(C) $y = -\frac{k}{3}x - \frac{15}{k}$ ✓

(D) $y = -\frac{k}{3}x + \frac{15}{k}$

1. Solve for y : $y = \frac{3}{k}x - \frac{15}{k}$. Slope of p is $\frac{3}{k}$ and the y -intercept is $\left(0, -\frac{15}{k}\right)$.

2. A perpendicular slope is the negative reciprocal: $-\frac{k}{3}$.

3. Line q keeps the intercept $-\frac{15}{k}$: $y = -\frac{k}{3}x - \frac{15}{k}$.

4. Check with $k = 3$: p has slope 1, q has slope -1 , both cross the y -axis at -5 .

Answer: (C)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x - 4)^2 + 125$. Which of the following statements is true?

(A) Only g attains a maximum, 125 ✓

(B) Neither attains a maximum

(C) Both f and g attain a maximum

(D) Only f attains a maximum, 120

1. g is a downward parabola with vertex $(4, 125)$ and $x = 4$ is in the domain, so g attains its maximum value 125.

2. $(0.8)^x > 0$ for every x , so $90(0.8)^x > 0$ and $f(x) < 120$ for every x . As x grows $f(x)$ gets closer to 120 but never equals it, so f has no maximum value even though it is bounded above by 120.

3. Only the third statement is correct.

Answer: (A)

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$ax + ay = 15$$

$$3ax - 2ay = 20$$

In the system of equations above, a is a nonzero constant. The ordered pair $(2, y)$ is a solution to the system. What is the value of a ?

1. Substitute $x = 2$: $2a + ay = 15$ and $6a - 2ay = 20$.
2. Treat the product ay as a single unknown P : $2a + P = 15$ and $6a - 2P = 20$.
3. Multiply the first by 2: $4a + 2P = 30$. Add to the second: $10a = 50$, so $a = 5$.
4. Check: $P = 15 - 10 = 5$, so $y = 1$; then $6(5) - 2(5) = 20$.

Answer: 5

6.

SAT CHALLENGE · CALCULATOR

In triangle ABC , angle C is a right angle and angle A measures 30° . The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?

(A) $10 + 10\sqrt{3}$

(B) $10\sqrt{3} - 10$

(C) $5 + 5\sqrt{3}$

(D) $15 + 5\sqrt{3}$ ✓

1. Let $BC = s$ (opposite 30°). Then $AC = s\sqrt{3}$ and $AB = 2s$.
2. Inradius of a right triangle: $r = \frac{BC + AC - AB}{2} = \frac{s + s\sqrt{3} - 2s}{2} = \frac{s(\sqrt{3} - 1)}{2} = 5$.
3. $s = \frac{10}{\sqrt{3} - 1} = \frac{10(\sqrt{3} + 1)}{2} = 5\sqrt{3} + 5$.
4. $AC = s\sqrt{3} = 5\sqrt{3}(\sqrt{3} + 1) = 15 + 5\sqrt{3}$.
5. (Area check: $\frac{1}{2} \cdot s \cdot s\sqrt{3}$ divided by the semiperimeter equals 5.)

Answer: (D)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$-2(x-5)^2 + k = m(x-1)(x-r)$$

In the equation above, k , m , and r are constants. The equation is true for all real values of x . What is the value of k ?

(A) -32 (B) 18 (C) 32 ✓(D) 50

1. Leading coefficients: left side $-2x^2 + \dots$, right side $mx^2 + \dots$, so $m = -2$.
2. The axis of the left side is $x = 5$; the right side has roots 1 and r , so $\frac{1+r}{2} = 5$ and $r = 9$.
3. Substitute the root $x = 1$, where the right side is 0: $-2(1-5)^2 + k = 0 \Rightarrow -32 + k = 0 \Rightarrow k = 32$.
4. (Check: $-2(x-5)^2 + 32 = -2x^2 + 20x - 18 = -2(x-1)(x-9)$.)

Answer: (C)

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all real values of x ,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k ?

1. Factor the base: $6^{2x+3} = 2^{2x+3} \cdot 3^{2x+3}$.
2. Divide by 2^{2x-1} : the powers of 2 give $2^{(2x+3)-(2x-1)} = 2^4 = 16$.
3. The powers of 3 give $3^{2x+3} = 3^3 \cdot 3^{2x} = 27 \cdot 9^x$.
4. So the left side equals $16 \cdot 27 \cdot 9^x = 432 \cdot 9^x$ and $k = 432$.

Answer: 432

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x + p)(x + q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?

- I. $p + q$ is an integer
- II. $p^2 + q^2$ is an integer
- III. $\frac{p^2 + q^2}{2}$ is an integer

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Expand: $x^2 + (p + q)x + pq$, so $p + q = -b$ and $pq = c$.
2. I: $p + q = -b$ is an integer. True.
3. II: $p^2 + q^2 = (p + q)^2 - 2pq = b^2 - 2c$, an integer. True.
4. III: $\frac{b^2 - 2c}{2}$ is an integer only when b^2 is even. Counterexample: $b = 3$, $c = 1$ gives $x^2 + 3x + 1$ with real roots $p, q = \frac{-3 \pm \sqrt{5}}{2}$, and $\frac{p^2 + q^2}{2} = \frac{9 - 2}{2} = \frac{7}{2}$. Not necessarily true.
5. Only I and II must be true.

Answer: (B)

10.

SAT CHALLENGE · CALCULATOR

$$(x - 1)^2 + (y + 3)^2 = 40$$

$$x^2 + y^2 - 4kx + 2ky = 9$$

In the xy -plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k ?

(A) $-1 + \sqrt{7}$

(B) $2 + \sqrt{7}$

(C) $1 + 2\sqrt{7}$

(D) $1 + \sqrt{7}$ ✓

1. The second circle has center $(2k, -k)$, since $x^2 - 4kx = (x - 2k)^2 - 4k^2$ and $y^2 + 2ky = (y + k)^2 - k^2$.

2. Substitute into the first equation: $(2k - 1)^2 + (-k + 3)^2 = 40$.

3. $4k^2 - 4k + 1 + k^2 - 6k + 9 = 40 \Rightarrow 5k^2 - 10k - 30 = 0 \Rightarrow k^2 - 2k - 6 = 0$.

4. $k = \frac{2 \pm \sqrt{4 + 24}}{2} = \frac{2 \pm 2\sqrt{7}}{2} = 1 \pm \sqrt{7}$.

5. Since $k > 0$, $k = 1 + \sqrt{7}$ (because $1 - \sqrt{7} < 0$).

Answer: (D)

11.

SAT CHALLENGE · CALCULATOR

Last year a shop sold only mugs and plates. This year the number of mugs sold rose by 30%, the number of plates sold fell by 10%, and the total number of items sold rose by 6%. Of the items sold this year, what percent were mugs, to the nearest tenth of a percent?

(A) 49.1% ✓

(B) 50.9%

(C) 52.0%

(D) 40.0%

1. Let last year's total be 100 with x mugs and $100 - x$ plates.

2. This year's total: $1.3x + 0.9(100 - x) = 106 \Rightarrow 0.4x = 16 \Rightarrow x = 40$.

3. This year there are $1.3(40) = 52$ mugs and $0.9(60) = 54$ plates, a total of 106.

4. Share of mugs = $\frac{52}{106} \approx 0.491$, i.e. 49.1%.

5. (Distractors: 40% is last year's share, 52% is this year's count out of 100, 50.9% is the plates' share.)

Answer: (A)

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a , b , and c are constants. What is the value of c ?

1. Since $x^3 - 1 = (x - 1)(x^2 + x + 1)$, multiply both sides by $x^3 - 1$: $a(x^2 + x + 1) - (bx + c)(x - 1) = 3x^2 + 8x + 4$.
2. Expand the subtracted product with its whole sign: $(bx + c)(x - 1) = bx^2 + (c - b)x - c$. So the left side is $(a - b)x^2 + (a - c + b)x + (a + c)$.
3. Match: x^2 : $a - b = 3$; x : $a + b - c = 8$; constant: $a + c = 4$.
4. From the constant equation $c = 4 - a$; from the first $b = a - 3$. Then $a + (a - 3) - (4 - a) = 8 \Rightarrow 3a = 15$, $a = 5$, so $b = 2$ and $c = -1$.

Answer: -1 **13.**

SAT CHALLENGE · CALCULATOR

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t ?

1. The inner radius is $10 - t$. Metal volume $= \pi(15) [10^2 - (10 - t)^2] = 765\pi$.
2. $100 - (10 - t)^2 = \frac{765}{15} = 51 \Rightarrow (10 - t)^2 = 49$.
3. $10 - t = 7$ or $10 - t = -7$, so $t = 3$ or $t = 17$.
4. Only $t = 3$ satisfies $0 < t < 10$.

Answer: 3

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A ?

1. Let W be the total downward force (table plus box) at first. Each of 4 legs carries $\frac{W}{4}$: $\frac{W}{4A} = 12 \Rightarrow W = 48A$.
2. Afterwards the force is $W + 120$, shared by 3 legs: $\frac{W + 120}{3A} = 24 \Rightarrow W + 120 = 72A$.
3. Substitute $W = 48A$: $48A + 120 = 72A \Rightarrow 24A = 120 \Rightarrow A = 5$.
4. Check: $W = 240$; $\frac{240}{20} = 12$ and $\frac{360}{15} = 24$.

Answer: 5

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

(A) 866

(B) 936 ✓

(C) 1160

(D) 616

1. The slant height of a face is the hypotenuse of a right triangle with legs 12 and half the base side that face sits on.
2. Slant height 13: half-side = $\sqrt{13^2 - 12^2} = 5$, so that face sits on a side of length 10. Slant height 20: half-side = $\sqrt{20^2 - 12^2} = 16$, so that face sits on a side of length 32.
3. Base area: $10 \cdot 32 = 320$.
4. Two faces with base 32 and slant height 13: $2 \cdot \frac{1}{2} \cdot 32 \cdot 13 = 416$. Two faces with base 10 and slant height 20: $2 \cdot \frac{1}{2} \cdot 10 \cdot 20 = 200$.
5. Total: $320 + 416 + 200 = 936$.

Answer: (B)

16.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, where D , E , and F are constants. The circle passes through the points $(2, 7)$ and $(10, 7)$, has radius 5, and has its center below the line $y = 7$. What is the value of $E + F$?

1. The points $(2, 7)$ and $(10, 7)$ have the same y -coordinate, so the center lies on the perpendicular bisector $x = 6$. The center is $\left(-\frac{D}{2}, -\frac{E}{2}\right)$, so $D = -12$.
2. Center $(6, k)$ with distance 5 to $(2, 7)$: $16 + (7 - k)^2 = 25 \Rightarrow (7 - k)^2 = 9 \Rightarrow k = 4$ or $k = 10$. Below $y = 7$ means $k = 4$.
3. $E = -2k = -8$. Also $F = h^2 + k^2 - r^2 = 36 + 16 - 25 = 27$.
4. $E + F = -8 + 27 = 19$.
5. (Check with $(2, 7)$: $4 + 49 - 24 - 56 + 27 = 0$.)

Answer: 19